Model Solution (Fall 2022)

University of Central Punjab

FOIT & CS

Mid-Term Exam

Program	BSCS	Semester	Fall 2022
Course Title	Multivariate Calculus	Course Code	CSSS 2733
Course Coordinator	Muhammad Rayees Ahmad	Time Allowed	90 minutes
Total Marks	40	Section	
Student Name		Registration No.	

Essay Type

Note. All computations should be part of the main work.

Q1. (7 marks) Consider $\vec{u} = 6\hat{i} - 5\hat{j} - 10\hat{k}$ and $\vec{v} = 5\hat{i} + 7\hat{j} - \hat{k}$, then

(i) Find $\vec{u} \cdot \vec{v}$

(ii) Find $\vec{u} \times \vec{v}$

(iii) Find $|\vec{u}|$ and $|\vec{v}|$

(iv) Find the angle θ between $\vec{u}$ and $\vec{v}$.

(**Marks distribution:** Dot product: 1 mark, cross product : 2 mark, finding $|\vec{u}|$: 1 mark, finding $|\vec{v}|$: 1 mark, finding angle θ: 2 marks).

Q2. (7 marks) Find an equation of the plane passing through the point $(-1, \frac{1}{2}, 3)$ and with normal vector $\hat{i} + 4\hat{j} + \hat{k}$. Also find intercepts.

(**Marks distribution:** Find x_0, y_0, z_0 and a, b, c: 1 mark, write formula for equation of plane: 1 mark, computations: 2 marks, write correct answer of equation of plane: 1 mark, find intercepts: 2 marks).

Q3. (10 marks) Find the distance from a point $S(3, 1, 5)$ to a line

$$L : \begin{cases} x = 3 - 2t \\ y = 4 + t \\ z = 6 - t \end{cases}$$

(**Marks distribution:** Find $\vec{v}$: 1 mark, find $|\vec{v}|$: 1 mark, find $\vec{PS}$: 1 mark, find $\vec{PS} \times \vec{v}$: 3 marks, find $|\vec{PS} \times \vec{v}|$: 1 mark, write formula for distance: 1 mark, computations: 1 mark, write correct answer: 1 mark).

Scanned with CamScanner

Q4. (4 marks) Find the point of intersection at which the line

$$L: \begin{cases} x = 1 + 2t \\ y = 4t \\ z = 2 - 3t \end{cases}$$

intersect the given plane $x + 2y - z + 1 = 0$.
(**Marks distribution:** Substitution of parametric equations of line in plane: 1 mark, compute value of t: 2 marks, find the point of intersection: 1 mark).

Q5. (6 marks) An object moves along a curve whose position vector is given by the vector function

$$\vec{r}(t) = \frac{1}{9-t}\,\hat{\imath} + \ln(t-5)\,\hat{\jmath} + \sqrt{t-1}\,\hat{k}$$

(i) Find the domain of $\vec{r}(t)$.
(ii) Find the velocity of the particle.
(**Marks distribution:** Finding domain : 3 marks, finding velocity: 3 marks).

Q6. (6 marks) Consider the table

Price

Income $I \setminus P$	3.00	3.50	4.00	4.50
20	2.65	2.59	2.51	2.43
40	4.14	4.05	3.94	3.88
60	5.11	5.00	4.97	4.84
80	5.35	5.29	5.19	5.07
100	5.79	5.77	5.60	5.53

shows the weekly beef consumption, C, (in lbs) of an average household as a function of p, the price of beef (in lbs) and I, annual household income (in $1000s$).

(a) What is the beef consumption of a house having annual income 80 (thousand dollars) buying beef at the price of 4 (dollars per pound)?
(b) At what price and index is the beef consumption at 4.05? Mention correct units.
(c) Give the table for beef consumption as a function of I with p fix at $p = 4.5$
(**Marks distribution:** Part (a) : 1 mark, part (b) : 2 marks, part (c) : 3 marks).

Scanned with CamScanner

MVC Midterm Model Solutions

Fall 2022

Q #1:

$$\vec{u} = 6\hat{i} - 5\hat{j} - 10\hat{k}$$

$$\vec{v} = 5\hat{i} + 7\hat{j} - \hat{k}$$

Sol:

(i) $\vec{u} \cdot \vec{v}$

$$\vec{u} \cdot \vec{v} = (6\hat{i} - 5\hat{j} - 10\hat{k}) \cdot (5\hat{i} + 7\hat{j} - \hat{k})$$

$$\vec{u} \cdot \vec{v} = (6)(5) + (-5)(7) + (-10)(-1)$$

$$\vec{u} \cdot \vec{v} = 30 - 35 + 10$$

$$\vec{u} \cdot \vec{v} = 40 - 35$$

$$\boxed{\vec{u} \cdot \vec{v} = 5}$$

ii) $\vec{u} \times \vec{v}$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 6 & -5 & -10 \\ 5 & 7 & -1 \end{vmatrix}$$

$$\vec{u} \times \vec{v} = \hat{i}\begin{vmatrix} -5 & -10 \\ 7 & -1 \end{vmatrix} - \hat{j}\begin{vmatrix} 6 & -10 \\ 5 & -1 \end{vmatrix} + \hat{k}\begin{vmatrix} 6 & -5 \\ 5 & 7 \end{vmatrix}$$

$$\vec{u} \times \vec{v} = \hat{i}(5 + 70) - \hat{j}(-6 + 50) + \hat{k}(42 + 25)$$

$$\boxed{\vec{u} \times \vec{v} = 75\hat{i} - 44\hat{j} + 67\hat{k}}$$

Scanned with CamScanner

iii) Find $|\vec{u}|$ and $|\vec{v}|$

$$\vec{u} = 6\hat{i} - 5\hat{j} - 10\hat{k} \quad , \quad \vec{v} = 5\hat{i} + 7\hat{j} - \hat{k}$$

$$|\vec{u}| = \sqrt{(6)^2 + (-5)^2 + (-10)^2} \quad , \quad |\vec{v}| = \sqrt{(5)^2 + (7)^2 + (-1)^2}$$

$$|\vec{u}| = \sqrt{36 + 25 + 100} \quad , \quad |\vec{v}| = \sqrt{25 + 49 + 1}$$

$$\boxed{|\vec{u}| = \sqrt{161}} \quad , \quad \boxed{|\vec{v}| = \sqrt{75}}$$

iv) Find θ

$$\cos\theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}|\,|\vec{v}|}$$

$$\theta = \cos^{-1}\left[\frac{\vec{u} \cdot \vec{v}}{|\vec{u}|\,|\vec{v}|}\right]$$

$$\theta = \cos^{-1}\left[\frac{(6\hat{i} - 5\hat{j} - 10\hat{k}) \cdot (5\hat{i} + 7\hat{j} - \hat{k})}{\sqrt{161}\,\sqrt{75}}\right]$$

$$\theta = \cos^{-1}\left[\frac{(6)(5) + (-5)(7) + (-10)(-1)}{\sqrt{161 \times 75}}\right]$$

$$\theta = \cos^{-1}\left[\frac{30 - 35 + 10}{\sqrt{12075}}\right] = \cos^{-1}\left[\frac{5}{\sqrt{12075}}\right]$$

$$\boxed{\theta = 87^\circ}$$

Scanned with CamScanner

Q #2

Given point $(-1, \frac{1}{2}, 3)$

This implies that

$$x_0 = -1, \quad y_0 = \frac{1}{2}, \quad z_0 = 3$$

Given normal vector: $\hat{i} + 4\hat{j} + \hat{k}$

This implies that:

$$a = 1, \quad b = 4, \quad c = 1$$

As we know that scalar equation of the plane is:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

putting values, we have

$$1(x - (-1)) + 4\left(y - \frac{1}{2}\right) + 1(z - 3) = 0$$

$$1(x + 1) + 4y - \frac{\cancel{4}^{2}}{\cancel{2}} + z - 3 = 0$$

$$x + 1 + 4y - 2 + z - 3 = 0$$

$$x + 4y + z + 1 - 2 - 3 = 0$$

$$x + 4y + z = 3 + 2 - 1$$

$$x + 4y + z = 5 - 1$$

$$\boxed{x + 4y + z = 4} \quad \text{—} (1)$$

This is the required equation of the plane.

<u>x-intercept</u>: put $y = 0 = z$ in eq (1), we get

$$x = 4$$

The point will be $(4, 0, 0)$

Scanned with CamScanner

y-intercept

put $x = 0 = z$ in eq (1), we get

$$4y = 4$$

$$\Rightarrow \boxed{y = 1}$$

The point will be $(0, 1, 0)$

z-intercept:

put $x = 0 = y$ in eq (1), we get

$$\Rightarrow \boxed{z = 4}$$

The point will be $(0, 0, 4)$.

Scanned with CamScanner

#3

Given point : $S(3,1,5)$

This implies that

$x_1 = 3 \quad , \quad y_1 = 1 \quad , \quad z_1 = 5$

Given line :

$$L : \begin{cases} x = 3-2t & \Rightarrow x = x_0 + at \Rightarrow x_0 = 3 \, , a = -2 \\ y = 4+t & \Rightarrow y = y_0 + bt \Rightarrow y_0 = 4 \, , b = 1 \\ z = 6-t & \Rightarrow z = z_0 + ct \Rightarrow z_0 = 6 \, , c = -1 \end{cases}$$

This implies that

$P_0(x_0, y_0, z_0) = P_0(3, 4, 6)$

and

$\vec{v} = \langle a, b, c \rangle = \langle -2, 1, -1 \rangle = -2\hat{i} + \hat{j} - \hat{k}$

Also, given point is $S(x_1, y_1, z_1) = S(3,1,5)$

So

$\overrightarrow{P_0S} = \overrightarrow{OS} - \overrightarrow{OP_0} = \langle 3, 1, 5 \rangle - \langle 3, 4, 6 \rangle$

$\overrightarrow{P_0S} = \langle 3-3 \, , \, 1-4 \, , \, 5-6 \rangle$

$$\boxed{\overrightarrow{P_0S} = \langle 0 \, , -3 \, , -1 \rangle}$$

Now

$$\overrightarrow{P_0S} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -3 & -1 \\ -2 & 1 & -1 \end{vmatrix} = \hat{i}\begin{vmatrix} -3 & -1 \\ 1 & -1 \end{vmatrix} - \hat{j}\begin{vmatrix} 0 & -1 \\ -2 & -1 \end{vmatrix} + \hat{k}\begin{vmatrix} 0 & -3 \\ -2 & 1 \end{vmatrix}$$

$$\overrightarrow{P_0S} \times \vec{v} = \hat{i}(3+1) - \hat{j}(0-2) + \hat{k}(0-6)$$

$$\boxed{\overrightarrow{P_0S} \times \vec{v} = 4\hat{i} + 2\hat{j} - 6\hat{k}}$$

Scanned with CamScanner

Taking magnitude

$$|\vec{P_0S} \times \vec{v}| = \sqrt{(4)^2 + (2)^2 + (-6)^2}$$

$$|\vec{P_0S} \times \vec{v}| = \sqrt{16 + 4 + 36}$$

$$\boxed{|\vec{P_0S} \times \vec{v}| = \sqrt{56}}$$

And

$$|\vec{v}| = \sqrt{(-2)^2 + (1)^2 + (-1)^2}$$

$$|\vec{v}| = \sqrt{4 + 1 + 1}$$

$$\boxed{|\vec{v}| = \sqrt{6}}$$

Formula :

$$S = \frac{|\vec{P_0S} \times \vec{v}|}{|\vec{v}|}$$

$$S = \frac{\sqrt{56}}{\sqrt{6}}$$

$$S = \sqrt{\frac{\cancel{56}^{28}}{\cancel{6}_{3}}}$$

$$S = \sqrt{\frac{28}{3}}$$

$$S = \sqrt{9.33}$$

$$\boxed{S = 3.055}$$

Scanned with CamScanner

Q#4

Given line:

$$L: \begin{cases} x = 1 + 2t \\ y = 4t \\ z = 2 - 3t \end{cases} \quad \text{--- (1)}$$

Given plane:

$$x + 2y - z + 1 = 0 \quad \text{--- (2)}$$

Using values of eq (1) in eq (2), we have

$$(1+2t) + 2(4t) - (2-3t) + 1 = 0$$

$$1 + 2t + 8t - 2 + 3t + 1 = 0$$

$$2t + 8t + 3t + 1 + 1 - 2 = 0$$

$$13t + \cancel{2} - \cancel{2} = 0$$

$$13t = 0$$

$$\Rightarrow \boxed{t = 0}$$

putting value of 't' in eq (1)

$$L: \begin{cases} x = 1 + 2t = 1 + 2(0) = 1 + 0 = 1 \Rightarrow \boxed{x = 1} \\ y = 4t = 4(0) = 0 \Rightarrow \boxed{y = 0} \\ z = 2 - 3t = 2 - 3(0) = 2 - 0 = 2 \Rightarrow \boxed{z = 2} \end{cases}$$

Hence the point of intersection is

$$(x, y, z) = (1, 0, 2)$$

Scanned with CamScanner

Q#5

$$\vec{r}(t) = \frac{1}{9-t}\,\hat{i} + \ln(t-5)\,\hat{j} + \sqrt{t-1}\,\hat{k}$$

Sol: (i) Domain of $\vec{r}(t)$

$f(t) = \frac{1}{9-t}$, $g(t) = \ln(t-5)$, $h(t) = \sqrt{t-1}$

Domain of $f(t)$

put $9-t=0$

$9=t$

$\boxed{t=9}$

This implies

$\text{Dom } f(t) = \mathbb{R} - \{9\}$

or

$\text{Domain } f(t) = (-\infty, 9) \cup (9, +\infty)$

Domain of $g(t)$

put $t-5>0$

$\Rightarrow t > 5$

This implies

$\text{Dom } g(t) = (5, +\infty)$

Domain of $h(t)$

put $t-1 \geq 0$

$\Rightarrow \boxed{t \geq 1}$

This implies

$\text{Dom } h(t) = [1, +\infty)$

Now

$$\text{Dom } \vec{r}(t) = \text{Dom } f(t) \cap \text{Dom } g(t) \cap \text{Dom } h(t)$$

$$\text{Dom } \vec{r}(t) = \big((-\infty, 9) \cup (9, +\infty)\big) \cap (5, +\infty) \cap [1, +\infty)$$

$$\text{Dom } \vec{r}(t) = \big((-\infty, 9) \cup (9, +\infty)\big) \cap (5, +\infty)$$

$$\boxed{\text{Dom } \vec{r}(t) = (5, 9) \cup (9, +\infty)}$$ Answer

Scanned with CamScanner

(ii) Find velocity

As $\vec{r}(t) = \frac{1}{9-t}\hat{i} + \ln(t-5)\hat{j} + \sqrt{t-1}\,\hat{k}$ —①

And we know that

$$\vec{v}(t) = \frac{d}{dt}\vec{r}(t)$$

So differentiate eq ① w.r.t 't'

$$\vec{v}(t) = \frac{d}{dt}\vec{r}(t) = \frac{d}{dt}\left[\frac{1}{9-t}\hat{i} + \ln(t-5)\hat{j} + \sqrt{t-1}\,\hat{k}\right]$$

$$\vec{v}(t) = \left[\frac{d}{dt}\left(\frac{1}{9-t}\right)\right]\hat{i} + \left[\frac{d}{dt}\ln(t-5)\right]\hat{j} + \left[\frac{d}{dt}(t-1)^{1/2}\right]\hat{k}$$

$$\vec{v}(t) = \left[\frac{(9-t)\cdot(0)-(1)(0-1)}{(9-t)^2}\right]\hat{i} + \left[\frac{1}{t-5}\cdot(1-0)\right]\hat{j} + \left[\frac{1}{2}(t-1)^{\frac{1}{2}-1}\cdot(1)\right]\hat{k}$$

$$\vec{v}(t) = \left[\frac{0+1}{(9-t)^2}\right]\hat{i} + \left[\frac{1}{t-5}\right]\hat{j} + \left[\frac{1}{2}(t-1)^{-1/2}\right]\hat{k}$$

$$\vec{v}(t) = \left[\frac{1}{(9-t)^2}\right]\hat{i} + \left[\frac{1}{t-5}\right]\hat{j} + \left[\frac{1}{2\sqrt{t-1}}\right]\hat{k}$$

Hence the required velocity vector is:

$$\boxed{\vec{v}(t) = \frac{1}{(9-t)^2}\hat{i} + \frac{1}{t-5}\hat{j} + \frac{1}{2\sqrt{t-1}}\hat{k}}$$

Scanned with CamScanner

#6

	I\p	Price 3.00	3.50	4.00	4.50
	20	2.65	2.59	2.51	2.43
	40	4.14	4.05	3.94	3.88
Income	60	5.11	5.00	4.97	4.84
	80	5.35	5.29	5.19	5.07
	100	5.79	5.77	5.60	5.53

C : Beef consumption (in lbs)

P : price of beef (in lbs)

I : Annual household income (in 1000s)

As

$$C = f(P, I)$$

(a) (Beef consumption of a house having annual income 80 buying beef at the price 4) = 5.19

(b) ~~At~~ The beef consumption is 4.05, at the annual household income of 40 (in 1000s) and 3.50 price (in lbs).

(c) $P = 4.50$

I\p	P=4.50
20	2.43
40	3.88
60	4.84
80	5.07
100	5.53

Scanned with CamScanner